Highlights

Transport-Inspired Rare-Event Sampling in Spin Systems: Learned Importance Improves Deep-Target Reliability

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- Adjoint importance and the rare-event committor share the same backward-conditioning structure and coincide for the unit-response absorbing-target problem studied here.
- Exact tests validate both the zero-variance importance limit and unbiased WE population control.
- At matched update cost and full replication ($R = 20$), WE holds an approximately two-fold precision advantage over naive sampling at the deepest resolved Ising-tail bin, though not uniformly across the tail.
- Learned importance improves deep-event discovery across six independent Edwards–Anderson disorder realizations (67.2% versus 52.8%; stratified odds ratio 2.33, $p = 0.0011$).
- The reliability improvement does not yet translate into wall-clock efficiency (0.33, 95% CI $[0.19, 0.58]$).

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ABSTRACT

Rare-event sampling requires computational effort to be concentrated in configurations that disproportionately determine a target probability. Radiation transport addresses this through adjoint importance functions and weight-window population control, and this work transfers that toolkit to rare-event sampling in lattice spin systems, using the shared backward-conditioning structure of the transport adjoint and the transition-path committor. A solvable birth–death process reproduces the zero-variance limit of exact importance, and enumeration of a 4×4 lattice confirms that weight-conserving Weighted Ensemble (WE) resampling is unbiased for any binning coordinate. At matched spin-update cost and full replication ($R = 20$), WE and naive sampling resolve the $L = 20$ Ising magnetization tail to comparable depth, with WE holding a modest, roughly two-fold precision advantage at the single deepest bin either method reaches. For the frustrated Edwards–Anderson spin glass, a self-consistency-trained neural coordinate acting on the full spin configuration bins the weighted population. Across six independent disorder realizations with one trained network each, the deepest calibrated target is reached in 121 of 180 replicas (67.2%) against 95/180 (52.8%) for the energy-binned coordinate. A disorder-stratified Cochran–Mantel–Haenszel analysis gives a common odds ratio of 2.33 (95% CI [1.40, 3.88], $p = 0.0011$); the same reliability advantage also appears within a single realization across three independently trained networks compared against a matched-size baseline (66.7% versus 48.9%, Fisher exact $p = 0.023$). That improvement is one of reliability rather than wall-clock efficiency. Repeated neural inference costs four to five times the wall-clock time of the energy coordinate, so the directly measured figure of merit sits at roughly one-third of parity (0.33, 95% CI [0.19, 0.58]). These results support the transfer of the tested transport-inspired framework and show that the reliability advantage extends beyond a single quenched-disorder realization. At the tested scale, neural inference overhead is the dominant measured barrier to wall-clock competitiveness.

1. Introduction

A rare event is one that direct simulation almost never observes: the nucleation of a new phase, a protein crossing between conformational basins [1], or a detector registering a particle that has penetrated many mean free paths of shielding [2, 3]. In each case the probability of the event is itself the quantity of scientific interest, and it is exactly that quantity which unbiased dynamics represents worst. If an event has equilibrium or first-passage probability $p \ll 1$, the relative error of a direct Bernoulli estimator formed from N effectively independent samples follows the elementary result [2, 3]

$$\varepsilon_{\mathrm{rel}} = \sqrt{\frac{1-p}{Np}} \simeq (Np)^{-1/2}, \qquad p \ll 1, \tag{1}$$

so the sample count required for fixed relative precision grows as $1/p$ and each additional decade of rarity costs an order of magnitude in computation. The failure is structural rather than incidental, and every established remedy responds to it the same way: not with more samples, but with better-placed ones. This elementary scaling is the origin of a broad family of rare-event techniques, including importance sampling, splitting, transition-path sampling, Forward Flux Sampling (FFS), Adaptive Multilevel Splitting (AMS), and Weighted Ensemble (WE) methods [1–3, 16, 20, 22, 38]. What each of them requires in practice is a rule for deciding which samples are the better-placed ones.

Two mature traditions answer that question using closely related mathematical objects. In transition-path theory, the committor $h(s)$ is the probability that a trajectory initiated at configuration s reaches a target set before a competing

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boundary, and it provides an optimal reaction coordinate for that event [4–6]. In classical particle transport, an adjoint solution assigns an importance to a phase-space point according to its expected contribution to a detector response; this importance can be used to bias sources, alter transition probabilities, and construct weight windows [7–11]. The formal connection to the Doob h-transform and conditioned Markov processes is well known [12–14]. The contribution of the present work is not to claim a new identity between these objects, but to carry out a systematic transfer and validation of the transport variance-reduction viewpoint in spin-system rare-event sampling, together with a controlled test of a learned configuration-space coordinate built on this viewpoint.

The transport perspective is attractive because it separates the ideal importance function from the population-control mechanism used when that importance is only approximate. Exact adjoint biasing can in principle produce zero-variance estimators, while weight windows use splitting and roulette to keep statistical weight near a prescribed range. This distinction underlies CADIS and related hybrid deterministic–Monte Carlo variance-reduction methods [10, 11, 15]. In the present setting, the same logic motivates using a committor-like quantity to organize a weighted population while retaining the original spin dynamics.

Rare-event population methods have a parallel history outside transport. WE was introduced for biomolecular transition sampling by Huber and Kim, shown to be statistically exact for broad classes of dynamics and binning procedures, and made practical for complex molecular systems by steady-state formulations and mature software [16–19]; FFS and AMS address related first-passage problems through interfaces or adaptive levels [20–22], while transition-path sampling works with reactive-path ensembles and can avoid prescribing a one-dimensional coordinate at all [1, 23]. Machine learning meanwhile offers several routes to the high-dimensional objects these methods need: flow-based and autoregressive samplers for equilibrium distributions [24–27], neural approximations of committor and Feynman–Kac observables from short trajectories [28, 29], learned bias potentials and importance functions [30, 31], and neural Doob dynamics for large deviations [32]. Together these motivate the central practical question of this paper: when the rare event occurs in a high-dimensional spin configuration space, how much is gained by replacing a hand-designed coordinate with a learned, committor-inspired one, measured against a strong physical baseline and with inference cost charged to the result?

The study is organized as a validation ladder built around three questions. Is the transport/committor correspondence mathematically valid in the formulated first-passage setting? How does weight-conserving WE alter sampling precision relative to direct Monte Carlo as the Ising tail becomes progressively rarer? And does a learned full-configuration coordinate improve deep-event access over a strong physical energy coordinate, reproducibly across independent instances of the problem rather than for one particular realization? A solvable one-dimensional birth–death chain first verifies the zero-variance limit of exact importance, exact enumeration of a 4×4 Ising lattice then verifies unbiased WE resampling, and a larger $L = 20$ Ising calculation tests the rarity-dependent precision trade-off at matched spin-update cost. The Edwards–Anderson (EA) spin glass finally provides a frustrated landscape in which magnetization is a poor coordinate and a low-energy target provides a demanding test for learned configuration-space importance [33–36]. Throughout, *reliability* (whether a method reaches the event at all, reproducibly) is distinguished from *efficiency* (variance reduction per unit of actual computational cost). The resulting picture is positive but bounded: WE is unbiased and gains precision only at the deepest resolved point of the Ising tail, while in the EA model the learned coordinate significantly improves deep-event discovery over energy across both independent neural initializations and six disorder realizations. That reliability gain does not translate into a wall-clock figure-of-merit gain once neural inference is charged, identifying inference overhead as the dominant measured barrier to computational competitiveness at the tested scale.

2. Theory and Methods

2.1. Adjoint importance, committors, and the zero-variance principle

Let $K(s, s')$ be the transition kernel of a Markov chain and let B denote a target set. For a first-passage event, the forward committor is the probability of reaching B before the competing set A [4, 5]:

$$h(s) = \Pr_s \left(\tau_B < \tau_A \right) . \tag{2}$$

For a discrete Markov process, this committor satisfies the backward (harmonic) equation in the interior of the state space, a standard result of transition-path theory for Markov jump processes [4–6]:

$$h(s) = \sum_{s'} K(s, s')\, h(s'), \qquad s \notin A \cup B, \tag{3}$$

with boundary values $h = 0$ on A and $h = 1$ on B. Conditioning the dynamics with a positive harmonic function leads to the Doob h-transform, introduced in classical potential theory [12] and used in this form throughout the large-deviation and conditioned-process literature [13, 14]:

$$K_h(s, s') = K(s, s')\,\frac{h(s')}{h(s)}, \tag{4}$$

where defined. This is the Markov-chain counterpart of adjoint-based importance biasing in transport. With the exact h, successful paths acquire identical total statistical weight, producing the familiar zero-variance ideal.

The relationship between the two importance functions should be stated precisely, since it is weaker than an identity. The committor of Eq. (2) is a probability: the chance of reaching B before A from s, bounded in $[0, 1]$ and defined only once a competing boundary is specified. The transport adjoint, by contrast, is an expected contribution to a specified detector response; it carries the units of that response and is unbounded in general. What the two share is structure rather than definition. Both are solutions of a backward equation generated by the same transition operator, both express the value of a state with respect to a terminal objective, and both play the same role when used to tilt or to bin. They coincide as functions in the particular case of an absorbing target with unit detector cross-section and no multiplication, which is the case constructed here, but they are not universally the same object and this paper does not claim they are. What transfers is the backward-conditioning structure and the machinery built on it, not an equivalence of the two quantities in general.

This correspondence motivates two distinct uses of approximate importance. The first changes transition probabilities directly, which can be fragile when h is inaccurate. The second retains the original dynamics but uses population splitting and merging to control statistical weight. The second strategy is adopted here using an unbiased, weight-conserving WE resampling scheme, so an imperfect binning coordinate can alter variance and sampling allocation without altering the estimator expectation [17].

The exact equivalence and zero-variance construction above apply to the formulated first-passage problem. In the equilibrium tail calculations below, the committor-inspired quantity is not used to claim an exact zero-variance transformation of the stationary distribution; it is used only as a progress coordinate for unbiased WE resampling. The first-passage theory therefore supplies the importance principle, while the spin-system calculations test its utility as a population-allocation coordinate.

The zero-variance statement can be seen directly at the path level. Consider a trajectory $\omega = (s_0, s_1, \ldots, s_\tau)$ generated under the Doob-transformed kernel until it reaches either A or B. Substituting the Doob transform of Eq. (4) into the path likelihood ratio gives the telescoping identity below, derived here for the discrete first-passage setting from Eqs. (3)–(4); it is the discrete-state counterpart of the classical zero-variance principle of transport importance sampling [7, 8, 11] and of the conditioned-process construction of Doob [12] and Chetrite and Touchette [14]:

$$\prod_{t=0}^{\tau-1} \frac{K(s_t, s_{t+1})}{K_h(s_t, s_{t+1})} = \prod_{t=0}^{\tau-1} \frac{h(s_t)}{h(s_{t+1})} = \frac{h(s_0)}{h(s_\tau)}. \tag{5}$$

On a successful path ending in B, $h(s_\tau) = 1$, so every successful history contributes exactly $h(s_0)$.

Fix an initial state s_0 with $h(s_0) > 0$ and let the target quantity be the first-passage probability $\pi = h(s_0) = \Pr_{s_0}(\tau_B < \tau_A)$. Generate M independent trajectories under K_h and assign each the likelihood-ratio weight of Eq. (5), which gives the following estimator, derived here:

$$\hat{\pi} = \frac{1}{M} \sum_{m=1}^{M} \left[\prod_{t=0}^{\tau_m-1} \frac{K(s_t^{(m)}, s_{t+1}^{(m)})}{K_h(s_t^{(m)}, s_{t+1}^{(m)})} \right] \mathbf{1}_B\!\left(s_{\tau_m}^{(m)}\right). \tag{6}$$

Here $\mathbf{1}_B(\cdot)$ denotes the indicator function of the target set B, equal to 1 if its argument lies in B and 0 otherwise. States with $h = 0$ receive zero transition probability, so the transformed chain never enters A and every trajectory terminates in B; on each such trajectory the weight telescopes to $h(s_0)$. Every term of the sum therefore equals $h(s_0)$, so $\hat{\pi} = \pi$ identically and $\mathrm{Var}(\hat{\pi}) = 0$ for any $M \geq 1$. This holds for the exact h; it degrades continuously, and in general non-uniformly, when h is replaced by an approximation. This is the property verified numerically in §3.1.

For an approximate importance function $\tilde{h}$, the exact telescoping property is no longer available. The present implementation separates importance representation from dynamical biasing instead: the physical Metropolis kernel is retained, while the learned or physical progress coordinate is used only to organize population splitting and merging. This design lets approximate importance information guide computational effort while weight conservation preserves the target measure.

Table 1 summarizes the operational correspondence used throughout the paper. The mapping is not intended to imply that all transport constructs carry over unchanged; source biasing, for example, has no literal analogue when the equilibrium spin distribution is fixed and the goal is to sample a stationary tail. The useful transfer is structural: an adjoint-like scalar identifies configurations that matter to the response, and a weight-management mechanism concentrates trajectory population in those configurations.

Radiation-transport concept	Markov/spin counterpart	Role in this study
Adjoint importance	Committor / event-conditioned value function	Defines ideal progress toward a rare target
Zero-variance importance bias	Doob h-transform / conditioned dynamics	Verified in the solvable first-passage demonstration
CADIS / weight-window construction	Importance-guided population allocation	Motivates the use of an importance coordinate to organize the weighted population
Weight windows	Weighted splitting and merging	Implemented through unbiased WE resampling with conserved statistical weight
Detector response	Rare-event indicator or threshold probability	Magnetization-tail or low-energy probability

Table 1: Transport concepts and their rare-event spin-system counterparts used in this work.

2.2. Spin models and rare-event targets

For the nearest-neighbor ferromagnetic Ising model, the Hamiltonian is that introduced by Ising [33] and solved exactly in two dimensions by Onsager [34],

$$H(\mathbf{s}) = -J \sum_{\langle ij \rangle} s_i s_j, \qquad s_i \in \{-1, +1\}, \tag{7}$$

with periodic boundaries and $J = 1$. The principal rare observable is the absolute magnetization $|m|$. For the spin glass, the corresponding random-bond Hamiltonian is that of Edwards and Anderson [35], in the $\pm J$ form reviewed by Binder and Young [36],

$$H(\mathbf{s}) = -\sum_{\langle ij \rangle} J_{ij} s_i s_j, \qquad J_{ij} \in \{-1, +1\}, \tag{8}$$

with one fixed disorder realization per comparison (six independent realizations are compared in §3.5). Quenched disorder and frustration make the low-energy states disordered, so $m \approx 0$ irrespective of energy and magnetization is not an adequate progress coordinate; low energy is used as the rare-event target instead. Production sampling uses checkerboard Metropolis sweeps [37] at $T = 2.6$ unless stated otherwise. The neural self-consistency objective uses a separate elementary single-spin Metropolis kernel, defined in §2.4, so that every one-spin neighbor can be enumerated exactly.

2.3. Weighted Ensemble sampling and rare-event calibration

WE propagates a weighted population under the original, unbiased Metropolis dynamics for a resampling interval τ. After propagation, walkers are partitioned into bins defined by a coordinate $q(s)$. If the total statistical weight in bin b is W_b, resampling redistributes that same W_b among the target number of walkers in the bin. Splitting creates additional trajectories with reduced individual weights this way, while merging removes redundant trajectories without changing total weight. The split/merge rule used here is an unbiased, weight-conserving WE resampling scheme [17]. Consequently, the coordinate controls where sampling effort is placed and can change estimator variance, while the expectation of the target estimator is preserved.

For a target set $\boldsymbol{B}$, WE estimates the target probability from the total statistical weight carried by walkers in $\boldsymbol{B}$. The estimator takes the weight-conserving form established for WE by Huber and Kim [16], proved statistically exact for broad classes of dynamics and binning procedures by Zhang et al. [17], and used in the steady-state and review formulations of Bhatt et al. [18] and Zuckerman and Chong [38]:

$$\hat{\pi}_t = \frac{\sum_i w_i^{(t)} \mathbf{1}_B\left(s_i^{(t)}\right)}{\sum_i w_i^{(t)}}, \tag{9}$$

Here i indexes the walkers in the population at resampling cycle t, $w_i^{(t)}$ and $s_i^{(t)}$ are the statistical weight and state of walker i, and $\mathbf{1}_B(\cdot)$ is the same indicator function of the target set B used in Eq. (6). The reported estimate is averaged over post-burn-in resampling cycles. Three coordinates are compared: absolute magnetization $|m|$, energy per spin E/N, and a learned scalar coordinate $I_\theta(s)$. Magnetization is deliberately retained as a mismatched baseline for the EA spin glass, whereas energy is a natural physical baseline for the low-energy target.

A fixed rare-event threshold can be either trivial or physically unreachable when lattice size, temperature, or disorder changes. The computational protocol uses a short pilot simulation instead, to locate the most extreme value reached under ordinary dynamics, and then places the production threshold beyond that pilot extreme by a small number of discrete lattice steps (spacing $2/L^2$ for magnetization, $4/L^2$ for EA energy per spin). The integer "beyond" records how many such steps the target is displaced past the pilot extreme; it provides an operational depth scale, without claiming that equal values of "beyond" correspond to identical probabilities across different system sizes or disorder realizations. The resampling interval is treated as a physical parameter rather than a bookkeeping choice. Independent diagnostics in the EA model show rapid relaxation of deep configurations, so shorter resampling intervals penetrate the deep ladder more effectively at comparable cost; the production deep-rare calculation uses $\tau = 2$ for this reason.

2.4. Neural importance coordinate

The learned coordinate is a compact periodic convolutional network acting on the full spin configuration. For a configuration s, the raw network output is denoted $\ell_\theta(s) = \log I_\theta(s)$. The production architecture applies circular padding, a 3×3 convolution from one input channel to 24 channels followed by ReLU, a second circularly padded 3×3 convolution from 24 to 24 channels followed by ReLU, and then global spatial mean and maximum pooling. The resulting 48 features are concatenated and mapped by a linear layer to the single unconstrained scalar $\ell_\theta(s)$. No sigmoid, exponentiation, clipping, normalization, or calibration is applied to that output before it is used for binning; the label WE[I_θ] is retained below as shorthand for WE binned on the raw log-importance coordinate ℓ_θ. The architecture is intentionally small: the question is whether a full-configuration coordinate can encode useful progress information that is absent from a scalar physical observable, not whether a large neural model can fit the training set.

The training objective is a self-consistency construction motivated by the harmonic backward equation and related to the label-free importance-learning strategy of Kim and Cai [31], adapted here to a single low-energy target and a bulk reference condition. This direction also sits within a broader neural-committor literature that includes direct variational approximation, semigroup and Feynman–Kac formulations, active importance sampling, and self-consistent committor-enhanced sampling [28, 29, 39–42]. For training only, an elementary one-spin Metropolis proposal kernel is used so that all $N = L^2$ neighbors can be evaluated. If $s^{(n)}$ denotes the state obtained by flipping site n, the kernel used in the loss is the standard Metropolis rule [37],

$$K_1\left(s, s^{(n)}\right) = \frac{1}{N} \min\left\{1, \exp\left(-\frac{\Delta E_n}{T}\right)\right\}, \qquad K_1(s,s) = 1 - \sum_{n=1}^{N} K_1\left(s, s^{(n)}\right). \tag{10}$$

For the $L = 16$ production calculations, all 256 one-spin neighbors are enumerated. If a neighbor lies in the target set, its boundary value is imposed exactly as $\ell_\theta(s^{(n)}) = 0$, corresponding to $I_\theta = 1$. For all other neighbors the network output is used. The harmonic update is evaluated in log space as

$$\log(K_1 I_\theta)(s) = \log\left[K_1(s,s) e^{\ell_\theta(s)} + \sum_{n=1}^{N} K_1\left(s, s^{(n)}\right) e^{\widetilde{\ell}_\theta(s^{(n)})}\right], \tag{11}$$

where $\widetilde{\ell}_\theta = 0$ for target neighbors and equals the network output otherwise. The implementation evaluates Eq. (11) with a log-sum-exp operation for numerical stability.

The target condition alone does not remove the constant solution of the harmonic equation: a constant I_θ satisfies $K_1 I_\theta = I_\theta$. A second, bulk-reference anchor is therefore imposed to make the learned coordinate non-degenerate. To avoid conflict with the first-passage sets A and B used in §2.1, let $\mathcal{T}$ denote target training states, $\mathcal{R}$ the bulk-reference states, and $\mathcal{Q}$ all non-target training states. The exact production loss, derived here for the present implementation from the harmonic fixed-point condition together with the two boundary constraints, is

$$\mathcal{L} = \mathcal{L}_{\mathrm{SC}} + \mathcal{L}_{\mathcal{T}} + \mathcal{L}_{\mathcal{R}}, \tag{12}$$

with

$$\begin{aligned} \mathcal{L}_{\mathrm{SC}} &= \frac{1}{|\mathcal{Q}|} \sum_{s\in\mathcal{Q}} \left[\log(K_1 I_\theta)(s) - \ell_\theta(s)\right]^2, \\ \mathcal{L}_{\mathcal{T}} &= \frac{1}{|\mathcal{T}|} \sum_{s\in\mathcal{T}} \ell_\theta(s)^2, \\ \mathcal{L}_{\mathcal{R}} &= \frac{1}{|\mathcal{R}|} \sum_{s\in\mathcal{R}} \left[\ell_\theta(s) + 6\right]^2. \end{aligned} \tag{13}$$

All three coefficients are exactly unity; no additional regularizer or weight-decay term is used. If a minibatch contains no target or no bulk-reference state, the corresponding boundary term is omitted for that minibatch. The bulk anchor is thus $\ell_{\mathcal{R}} = -6$ ($I_{\mathcal{R}} = e^{-6} \simeq 2.48\times10^{-3}$). Its numerical value sets a reference scale and breaks the constant-solution degeneracy; it is not asserted to be an exact committor boundary value.

Training configurations for the six-disorder production study are collected with a separate WE-style coverage procedure driven by the oriented physical progress coordinate $-E/N$. The collector starts from $80 \times 40 = 3200$ configurations, performs one checkerboard Metropolis sweep per collection iteration for 3000 iterations, partitions the observed progress range into 12 adaptive bins, equalizes occupancy across occupied bins, and retains a population snapshot every third iteration. A boundary anchor is either already in the target or one single-spin flip away from it. If the first collection contains no such anchor, one retry uses 4500 iterations and 6400 initial configurations. The final training subset contains at most $n_{\mathrm{train}} = 2500$ states. Boundary-adjacent states are capped at $0.70 n_{\mathrm{train}} = 1750$, ensuring that non-boundary states remain available for the second boundary condition. For the EA model, bulk-reference states are selected on the bulk side of the pilot energy ladder; equivalently, they satisfy $E/N \geq e_{\mathrm{bulk}}$, where e_{bulk} is the pilot mean energy per spin plus two pilot standard deviations.

Production training uses Adam with its default momentum parameters, learning rate 2×10^{-3}, mini-batches of 128, and 20 epochs, with both the NumPy collection seed and PyTorch training seed fixed to 0 for the one-network-per-disorder comparison. The single-spin kernel in Eqs. (10)–(11) is used only to define the self-consistency objective; production trajectories themselves use the checkerboard Metropolis sweeps described in §2.2. This operator mismatch can affect the quality of the learned coordinate, but not estimator correctness, because ℓ_θ enters only through WE binning rather than through a likelihood-ratio tilt or modified transition probability. Accordingly, I_θ is termed a *learned importance coordinate*, not a neural committor: no quantitative committor accuracy is claimed or required for the unbiased WE estimator [17].

2.5. Computational cost and statistical analysis

Two cost measures are used. Matched-cost comparisons of sampling reach use the number of attempted spin updates as the common Monte Carlo work unit, natural for the Ising-tail comparison since both direct sampling and WE use the same Metropolis kernel. That measure is blind to neural inference, though, which is performed repeatedly during WE binning; wall-clock time is also recorded directly for this reason, and is the measure used whenever a figure of merit is quoted for a neural coordinate. The figure of merit is the cost-normalized inverse variance standard in transport variance-reduction studies [8, 15],

$$\mathrm{FOM} = \frac{1}{\sigma_{\mathrm{rel}}^2 T}, \tag{14}$$

with T the computational cost charged to the calculation.

Suppose R independent replicas produce estimates $\hat{\pi}_r$. Following the independent-replica uncertainty reporting recommended for WE studies [38] and implemented in WESTPA [19], the across-replica relative standard deviation is

$$\sigma_{\mathrm{rel}} = \frac{\mathrm{sd}(\hat{\pi}_1, \ldots, \hat{\pi}_R)}{\bar{\pi}}, \qquad \bar{\pi} = \frac{1}{R}\sum_{r=1}^{R} \hat{\pi}_r, \tag{15}$$

but for deep targets the most informative quantity can instead be the fraction of independent runs that observe the event at all, so zero-hit counts are reported explicitly. Event-positive and zero-hit replicas are compared using the exact conditional test for a 2×2 contingency table introduced by Fisher [44], appropriate here because the expected counts are small. For the six-disorder comparison, the descriptive pooled Fisher test is complemented by a Cochran–Mantel–Haenszel analysis that preserves disorder realization as a stratum and estimates a common odds ratio [43]. Confidence intervals on figure-of-merit ratios are obtained by the nonparametric bootstrap [45] in a two-level form that resamples the trained-network (or disorder-realization) axis in the outer loop and the replica axis in the inner loop.

During software development, ChatGPT (OpenAI) and Claude (Anthropic) were used as coding assistants for code drafting, refactoring, and diagnostic review. Production calculations were executed under the recorded configurations, and code changes affecting reported results were checked against the exact validation gates described in §3.1.

3. Results

The numerical study proceeds from exact verification to progressively harder sampling tests: first the zero-variance and unbiasedness gates, then the Ising rare-tail benchmark, and finally the calibrated deep-rare EA comparisons used to assess learned importance.

3.1. Exact validation of importance sampling and WE

A biased one-dimensional birth–death chain provides a solvable test. Starting at $s_0 = 1$ with absorbing boundaries at 0 and $N = 15$ and step probability $p = 0.4$ toward the rare boundary, the exact committor is $h(s_0) = 1.144443{\times}10^{-3}$. Direct Monte Carlo gives a relative variance of 2.17×10^{-3} for 4×10^5 histories, in agreement with the Bernoulli prediction $1/(Np)$. Using the exact importance transform reduces the measured relative variance to approximately 10^{-30}, numerically reproducing the zero-variance prediction of Eq. (5).

The second exact gate tests the weighted-population implementation itself. For the 4×4 Ising model, all 2^{16} configurations can be enumerated. Across three validation regimes (above T_c, below T_c with the double-well structure, and the folded $|m|$ distribution), the mean $|\Delta \ln P|$ between WE and exact enumeration ranges from approximately 0.006 to 0.024; a separate learned-importance gate gives target-probability ratios of 1.000 and 1.001 relative to exact enumeration when WE is binned by magnetization and energy, respectively. These tests confirm that the implemented WE resampling reproduces the exact target measure under distinct binning coordinates, so later coordinate comparisons are interpreted as efficiency or reliability differences rather than changes in the target expectation.

Test	Reference	Outcome
Birth–death importance transform	analytic committor	exact-importance relative variance $\sim 10^{-30}$
4×4 Ising WE distribution	exact enumeration	mean $\lvert\Delta \ln P\rvert$ = 0.006–0.024 across three regimes
4×4 learned-coordinate gate	exact target probability	WE ratios 1.000 and 1.001

Table 2: Validation ladder used before interpreting the larger rare-event calculations.

3.2. Rarity-dependent performance in the Ising ferromagnet

At $L = 20$ and matched update cost ($\sim 3.4 \times 10^8$ spin updates), naive sampling and WE agree closely through the bulk and into the near tail, with $R = 20$ independent replicas per method. At the second-deepest resolved bin ($|m| = 0.886$), naive sampling is in fact the more precise of the two at matched cost: relative standard deviation 0.36 against WE's 0.79, a figure-of-merit advantage for naive of roughly 5×. Only at the single deepest bin reached by either method ($|m| = 0.932$, $P \approx 1.1 \times 10^{-7}$ for WE against $P \approx 7.4 \times 10^{-8}$ for naive) does WE hold an edge, with a figure of merit about 2× that of naive (relative standard deviation 3.10 versus 4.47); both estimates are individually noisy at this depth.

The comparison therefore shows a rarity-dependent crossover in precision rather than a uniform algorithmic advantage. Direct sampling is more efficient in the near tail, where the event is still observed often enough that population control adds overhead, whereas WE becomes more precise at the deepest resolved bin, where direct-sampling variance has grown sharply. The exact gates of §3.1, rather than this performance comparison, establish the correctness of the WE implementation.

3.3. Calibration of the deep-rare EA regime

Having established the rarity-dependent behavior of WE in the ferromagnetic benchmark, the analysis next moves to the frustrated EA model, where the principal learned-coordinate test is placed deliberately beyond the range where direct sampling is consistently informative. Table 3 summarizes the depth calibration at $L = 16$ and identifies the access-dominated regime used for the learned-coordinate test.

Beyond	$\hat{\pi}$	relative s.d.	zero-hit replicas
1	5.29×10^{-5}	0.23	0/10
2	5.48×10^{-6}	0.46	0/10
3	7.14×10^{-7}	1.61	7/10
4	0	—	10/10

Table 3: Depth calibration for the EA model at $L = 16$ using 10 independent direct-sampling replicas. The integer "beyond" indexes successively deeper discrete energy thresholds.

The beyond= 4 threshold is used as a stress test of rare-event access for this reason: direct Monte Carlo is fully blind there, so the learned-coordinate comparison directly tests whether a configuration-space importance map can organize the weighted population so that trajectories continue to enter a region essentially absent from the direct Monte Carlo sample.

At beyond= 4, direct Monte Carlo is effectively blind under the calibrated budget. The WE budget was increased to 80 bins, 40 walkers per occupied bin, 3000 iterations, and $\tau = 2$.

3.4. Learned importance within one disorder realization

Three independently trained self-consistency networks were compared with the same physical-coordinate baselines on one EA disorder realization ($L = 16$, $T = 2.6$), 30 replicas per method. Using independently trained networks establishes a reproducible coordinate-level effect rather than a favorable outcome from one particular initialization.

Method	event-positive replicas	discovery rate
naive	1/30	3.3%
WE[m]	5/30	16.7%
WE[E] (balanced, $n = 90$)	44/90	48.9%
WE[I_θ], seed 0	19/30	63.3%
WE[I_θ], seed 1	20/30	66.7%
WE[I_θ], seed 2	21/30	70.0%
WE[I_θ], pooled ($n = 90$)	60/90	66.7%

Table 4: Deep-rare EA comparison at $L = 16$, beyond= 4, one disorder realization. Each learned-coordinate row is a separately trained network evaluated over 30 replicas.

The three independently trained networks give discovery rates of 63.3%, 66.7%, and 70.0%. An expanded independent WE[E] baseline of 90 replicas is used for the balanced comparison, giving $44/90$ (48.9%) against the pooled learned-coordinate result of $60/90$ (66.7%). Fisher's exact test on this balanced comparison gives

$$\mathrm{WE}[I_\theta] : \ 60/90\,(66.7\%) \quad \text{versus} \quad \mathrm{WE}[E] : \ 44/90\,(48.9\%), \qquad p = 0.023. \tag{16}$$

This is the central within-disorder result: the learned importance map significantly increases the probability that a finite-budget WE run reaches the deep target, reproducibly across three independent initializations.

The comparison is informative because energy is a strong physical baseline for this task: the target itself is defined by low energy, so an improvement over WE[E] is not a comparison against an irrelevant coordinate. The network receives the full spin configuration and can distinguish configurations with similar instantaneous energy but different local structure and different dynamical prospects of progressing toward the target. Two configurations at identical energy can have very different committor values on a frustrated landscape, after all, and an energy-adjacent label cannot represent that difference. The role of the learned function remains conservative throughout: it defines the WE partition rather than replacing the physical transition kernel, so the learned model influences how computational effort is distributed across configuration space while the unbiased WE resampling rule preserves the target expectation.

3.5. Generalization across disorder realizations

The result above establishes reproducibility across network initializations on one J_{ij} sample; it does not by itself show that the effect extends beyond that particular disorder realization. Five further independent EA disorder realizations were therefore tested at $L = 16$ and $T = 2.6$ using the same beyond= 4 calibration rule, one trained network each. The absolute target threshold is disorder-specific because it is placed four discrete energy steps beyond that realization's pilot extreme; the resulting $e_* = E^*/N$ values are reported in Table 5. Realization 12345 retains its previously calibrated explicit value, whereas the other five thresholds are calibrated independently with the same pilot protocol. Because the within-disorder experiment already probes initialization robustness, the additional computational budget was allocated to disorder variation rather than to another three-network ensemble for each realization.

Disorder	e_*	WE[I_θ]	WE[E]	Fisher p
12345 (original)	−1.1190	19/30 (63.3%)	12/30 (40.0%)	0.120
23456	−1.1328	15/30 (50.0%)	8/30 (26.7%)	0.110
45678	−1.1016	29/30 (96.7%)	26/30 (86.7%)	0.353
56789	−1.0938	20/30 (66.7%)	15/30 (50.0%)	0.295
67890	−1.1172	28/30 (93.3%)	29/30 (96.7%)	1.000
78901	−1.1641	10/30 (33.3%)	5/30 (16.7%)	0.233
Pooled (6 realizations)	–	121/180 (67.2%)	95/180 (52.8%)	0.0071

Table 5: Deep-rare EA comparison repeated across six independent disorder realizations, one trained network each, beyond= 4, 30 replicas per method. The target threshold $e_* = E^*/N$ is calibrated separately for each realization. Realization 12345 is the single-network reading of Table 4 (seed 0), not the three-network pooled result.

No single realization reaches significance on its own at $n = 30$. Realizations 45678 and 67890 are near ceiling for both methods, which limits their discriminating power; realization 67890 is also the only stratum in which WE[E] has the nominally higher discovery rate (96.7% versus 93.3%). In the other five of six realizations the learned coordinate has the higher discovery rate. For descriptive comparison, pooling all six realizations gives:

$$\text{WE}[I_\theta]:\ 121/180\,(67.2\%) \quad \text{versus} \quad \text{WE}[E]:\ 95/180\,(52.8\%), \qquad p = 0.0071. \tag{17}$$

Because the six comparisons have different disorder-specific event rates, inferential analysis also preserves disorder as a stratum. The Cochran–Mantel–Haenszel analysis gives a common odds ratio of 2.33 (95% CI [1.40, 3.88]) with $p = 0.0011$ [43], corroborating the descriptive pooled result without treating the six disorder realizations as one homogeneous sample. Together with the within-disorder network replication, this shows that the reliability advantage extends beyond a single quenched-disorder realization and is reproduced across the six realizations tested. Wall-clock cost ratios for the five additional realizations (4.45–4.97 relative to WE[E]) are consistent with the other measurements in this study (§4.2), so the cost penalty is likewise reproduced across disorder.

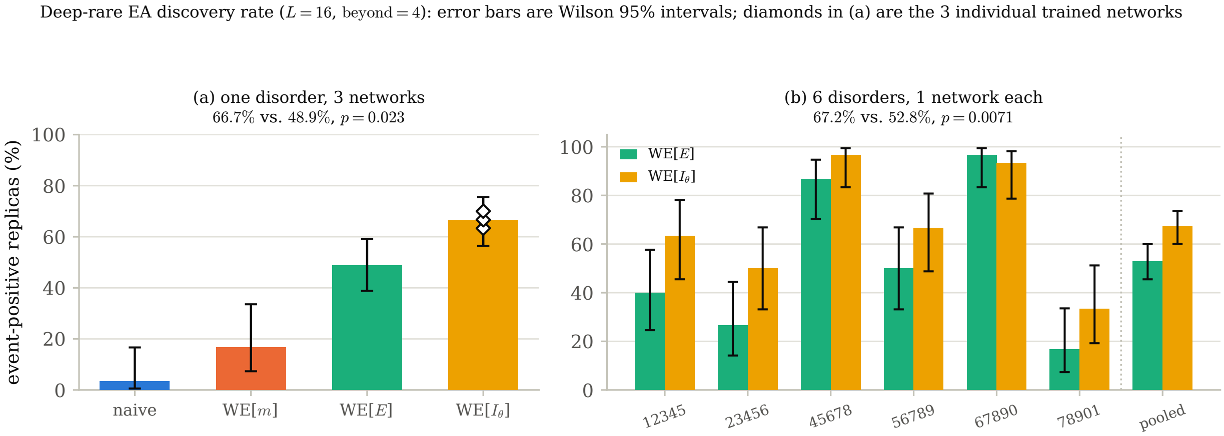

Figure 1: The learned importance coordinate reaches the deep target more reliably than the physical energy coordinate. Error bars are Wilson 95% intervals. (a) Discovery rate at beyond= 4 (disorder realization 12345): open diamonds are the three independently trained networks (Table 4); WE[E] uses the balanced 90-replica baseline. (b) The same comparison repeated across six independent disorder realizations, one trained network each (Table 5), with the pooled result at right.

4. Discussion

4.1. From transport importance to learned deep-event reliability

The results support a hierarchical transfer of transport-inspired variance-reduction ideas to rare-event sampling in spin systems. At the theoretical level, the committor supplies an event-conditioned importance function for the first-passage problem, while the exact Doob transform provides the zero-variance reference. The birth–death calculation verifies that limit directly. At the algorithmic level, the exact-enumeration gates show that the unbiased, weight-conserving WE resampling construction can exploit an approximate progress coordinate without changing the target expectation. This separation between correctness and coordinate quality is essential for the learned-coordinate calculations: an imperfect coordinate may waste computational effort, but it does not redefine the probability being estimated.

The $L = 20$ Ising calculation shows how the value of population control changes with rarity. Direct sampling is more efficient in the better-populated part of the tail, whereas WE becomes more precise at the deepest resolved bin under the same spin-update budget. The comparison therefore does not support a uniform advantage for WE across the full tail; instead it shows the rarity-dependent crossover expected of a variance-reduction method whose population-control overhead becomes worthwhile only after ordinary sampling grows sufficiently sparse. This observation provides a useful bridge between the exact validation tests and the substantially deeper EA calculation, where the central difficulty is no longer modest variance reduction but whether an independent run reaches the target at all.

This progression leads to the central empirical result in the frustrated EA model. Energy is already a strong physical coordinate for the selected low-energy event, so WE[E] is a demanding baseline rather than an intentionally weak comparator. Nevertheless, the full-configuration coordinate I_θ increases the frequency with which independent WE calculations reach the calibrated deep target. The effect is reproduced along two distinct axes. On one disorder realization, three independently trained networks give discovery rates of 63.3%, 66.7%, and 70.0% against a balanced 48.9% energy-coordinate baseline. Across six independent quenched-disorder realizations, one trained network per realization gives an aggregate discovery rate of 67.2% for WE[I_θ] versus 52.8% for WE[E].

Because the absolute event rate varies from one disorder realization to another, the stratified analysis is more informative than treating all replicas as one homogeneous sample. The Cochran–Mantel–Haenszel estimate gives a common odds ratio of 2.33 (95% CI $[1.40, 3.88]$, $p = 0.0011$), consistent with the descriptive pooled comparison and with the fact that five of the six disorder realizations favor the learned coordinate. The near-ceiling realizations contribute relatively little discriminating information, but they do not reverse the overall direction of the stratified effect. Taken together, the results show that the reliability advantage is not confined to a particular neural initialization or a single quenched-disorder sample; it is reproduced across the six disorder realizations tested.

The physical interpretation is that energy, although informative, compresses a frustrated spin configuration to one scalar. Configurations with similar instantaneous energy can differ in local frustration pattern and in their dynamical prospects of progressing toward the target. A full-configuration importance coordinate can therefore encode progress information that is not represented by energy alone. The present results establish the utility of that information for WE population allocation; they do not require, and do not claim, that I_θ is a quantitatively exact committor.

4.2. Reliability versus computational efficiency

Improved discovery reliability does not, by itself, imply a computational speedup. On the spin-update cost measure, the three within-disorder trained networks give figure-of-merit ratios of 1.29, 1.07, and 1.04 against WE[E], with a two-level-bootstrap geometric-mean ratio of 1.16 (95% CI [0.65, 2.12]). That measure is incomplete for a neural coordinate because it does not charge the repeated network evaluations performed at every resampling step.

Wall-clock time was therefore recorded directly for the final deep-rare calculations. Across the tested networks and disorder realizations, WE[I_θ] required approximately four to five times the wall-clock time of WE[E]. Charging both methods on a like-for-like native replica basis gives

$$\mathrm{FOM}_{\mathrm{wall}}(\mathrm{WE}[I_\theta])\big/\mathrm{FOM}_{\mathrm{wall}}(\mathrm{WE}[E]) = 0.33, \qquad 95\%\ \mathrm{CI} = [0.19, 0.58], \tag{18}$$

which lies below parity. The distinction is therefore substantive rather than semantic: the learned coordinate organizes the weighted population more effectively for deep-target access, but the present CPU implementation pays enough inference overhead that this reliability gain does not translate into higher wall-clock efficiency. At the tested scale, repeated neural inference is the dominant measured limitation on computational competitiveness. Reducing that overhead through cheaper architectures, caching, batching, or hardware acceleration is a natural target for a dedicated efficiency study.

4.3. Relation to existing methods, scope, and outlook

The present implementation is one member of a broader splitting and path-sampling family. FFS and AMS advance trajectories through fixed interfaces or adaptively chosen levels [20–22], and transition-path sampling works directly with reactive-path ensembles, which frees it from a fixed scalar coordinate but targets different estimators than the stationary tail probabilities studied here [1, 23, 46]. WE differs in maintaining a continuously weighted population and resampling within bins, so that the bins influence variance but not the target measure [16, 17, 38]. What the transport analogy adds is a principled way to think about where that weighted population should be concentrated: CADIS couples deterministic importance information to source biasing and weight-window construction [10, 11, 15], while the present implementation adopts the population-control part of that logic without altering the Metropolis kernel.

Against the neural rare-event literature, the principal distinction is where the learned object enters the algorithm. Flow-based and autoregressive samplers learn global equilibrium structure [24–27]; neural committor and Feynman–Kac methods target dynamical progress toward a specified event [28, 29, 39]; and active or self-consistent committor schemes concentrate training and sampling effort in transition-relevant regions [40–42]. The framework of Kim and Cai is closest in spirit, combining a learned event-specific importance function with biased transitions and branching [31]. In the notation of §2.1, their learned importance plays the role of h, their modified transitions the role of the tilt, and their branching the role of splitting. The present study instead inserts the learned function only in the WE partition. This more conservative use of learned importance preserves the original physical dynamics and allows coordinate quality to be assessed without introducing likelihood-ratio error from an approximate directional bias.

Within this methodological context, the demonstrated scope can be stated precisely. The principal learned-coordinate experiment concerns two-dimensional EA systems at $L = 16, T = 2.6$, and the calibrated beyond= 4 target, using six independent disorder realizations and a compact periodic convolutional architecture. Within that scope, the reliability advantage is reproduced across quenched disorder and across independent neural initializations. The present data do not determine how the magnitude of that advantage changes with lattice size, temperature, or a substantially larger disorder ensemble, and I_θ is used as a committor-inspired importance coordinate rather than as a quantitatively validated estimate of the exact committor. These boundaries do not alter the demonstrated result; they define the next tests needed to establish broader scaling and mechanistic interpretation.

Natural extensions are therefore to repeat the deep-target comparison across L and T, to benchmark learned coordinates against tractable reference committors on smaller systems, and to test architectures that encode spin symmetry or local frustration more directly. From a computational perspective, reducing inference overhead is equally important, because the current results show that better deep-event reliability and better wall-clock efficiency are separate objectives. The transport-inspired framework is compatible with both directions: coordinate quality can be improved without changing the unbiased WE estimator, while the population-control machinery can remain fixed as model and hardware choices evolve.

5. Conclusions

Adjoint transport theory provides a useful organizing principle for rare-event sampling in lattice spin systems. The committor supplies the event-specific importance function, the exact Doob transform defines the zero-variance ideal, and weight-conserving population control provides a practical way to exploit approximate importance without changing the target measure; these elements were validated sequentially from a solvable birth–death process to exact enumeration of the Ising model.

At system scale, WE and naive sampling resolve the $L = 20$ Ising magnetization tail to comparable depth at $R = 20$. Direct sampling is more precise in the near tail, while WE holds a modest, roughly two-fold figure-of-merit advantage at the single deepest resolved bin. The benefit of population control is therefore rarity-dependent rather than uniform across the tail.

In the frustrated EA model, the learned configuration-space importance coordinate produces the main new result of this study. Across six independent disorder realizations with one trained network each, it reaches the deep target in $121/180$ replicas against $95/180$ for the physical energy coordinate (67.2% versus 52.8%). Preserving disorder realization as a stratum gives a common odds ratio of 2.33 (95% CI $[1.40, 3.88]$, $p = 0.0011$), while the descriptive pooled Fisher comparison gives $p = 0.0071$. The same direction is reproduced within one disorder across three independently trained networks against a balanced baseline (66.7% versus 48.9%, $p = 0.023$). A learned progress coordinate can extract dynamical information not fully represented by a scalar physical observable, and the advantage is not confined to one particular quenched-disorder sample: it is reproduced across the six disorder realizations tested.

That advantage is presently one of reliability rather than efficiency: repeated network inference is directly measured to cost four to five times WE$[E]$'s wall-clock time, leaving the learned coordinate at roughly one-third of parity in figure of merit (0.33, 95% CI $[0.19, 0.58]$). At the tested scale, repeated neural inference is therefore the dominant measured limitation on wall-clock efficiency.

Data and Code Availability

The code, input data, trained network weights, quenched disorder realizations, per-replica outputs, validation data, and wall-clock records required to reproduce the reported results have been assembled into an archival package. A persistent public repository DOI will be inserted before final submission. The package also records the software environment, configuration files, and random seeds used for the production calculations so that the reported analyses can be reproduced under the archived environment.

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CRediT authorship contribution statement

M.Y. Alzaatreh: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT (OpenAI) and Claude (Anthropic) to assist with manuscript organization, language refinement, critical review, and software support. After using these tools, the author reviewed and edited the content as needed, independently verified the scientific claims and numerical results, and takes full responsibility for the content of the publication.

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